Climate Policy

ISSN: 1469-3062 (Print) 1752-7457 (Online) Journal homepage: www.tandfonline.com/journals/tcpo20

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To cite this article: Chirag Dhara, Anshuman Gupta, Ishita Bagri, Sebastián Block & Kamal Kumar Murari (22 Oct 2025): Interpreting climate performance indices: implications for equitable and effective policy, Climate Policy, DOI: 10.1080/14693062.2025.2574858

To link to this article: https://doi.org/10.1080/14693062.2025.2574858

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Published online: 22 Oct 2025.

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CLIMATE POLICY
https://doi.org/10.1080/14693062.2025.2574858

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Interpreting climate performance indices: implications for equitable and effective policy

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ABSTRACT

Climate performance indices play a crucial role in evaluating countries' efforts and advancing global environmental governance. This study critically examines disparities among prominent climate performance indices, including the Environmental Performance Index (EPI), Climate Change Performance Index (CCPI), and Climate Action Tracker (CAT). Our analysis reveals significant divergences in country rankings, particularly between 'developed' and 'least developed' nations, underscoring how subjective methodological choices impact indices' results and their interpretation. We develop an analytic tool, called EPI-equity, to demonstrate how integrating equity principles can substantially alter performance assessments. We furthermore propose a novel conceptual framework to classify indices based on the choice of methodological framework and embedded normative choices, highlighting how these can shape the interpretation of performance. These results allow the contextualization of outcomes of climate performance indices and the degree to which they concur with one another. Thus, this framework helps translate methodological choices into a conceptual understanding of what an index truly measures. We propose that explicitly articulating the normative choices embodied by performance indices can enhance transparency, guide developers in aligning methodological choices with intended interpretations, and equip users with a clearer understanding of the results. Our analysis highlights the importance of employing multiple indices that encompass a range of normative choices for a comprehensive evaluation of countries' climate performance. This adaptable framework provides a structured approach to guide the selection of indices spanning a broad spectrum of viewpoints, and, thereby, mitigates the likelihood of conflicts arising from fragmented worldviews on complex socio-environmental issues.

ARTICLE HISTORY
Received 4 December 2024
Accepted 8 October 2025

KEYWORDS
Climate performance indicators; environmental performance index; climate action tracker; equity; fairness

Key Policy Insights

- Climate performance indices can yield contradictory assessments due to differing underlying approach of normative choices.
- No single index can capture all dimensions of climate performance. Policymakers should use multiple indices to guide decision-making, ensuring a comprehensive evaluation of national climate performance.
- Policymakers benefit when index developers explicitly articulate both the methodological framework and normative preferences embedded in their normative frameworks. This transparency helps mitigate conflicts such as those seen between the CCPI and EPI developers and the Australian and Indian governments, respectively.

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/14693062.2025.2574858.

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1. Introduction

Environmental indices help interpret complex multi-dimensional information, monitor progress toward sustainability goals (Meadows, 1998; Parris & Kates, 2003; Singh et al., 2012), and advance environmental governance (DeLoyde & Mabee, 2020; UNCED, 1992). Various indices such as the Sustainable Development Goals Index (Sachs et al., 2016), Ecological Footprint (Wackernagel et al., 1999), and the Comprehensive Environmental Performance Index (Latif, 2022) are designed to measure countries' environmental performance, informing policymakers, international agencies, researchers, and the public. Careful design of indices is critical for contextualization, data quality, and transparent methodologies (Nardo et al., 2008). Additionally, indices must be comprehensive and multi-dimensional (Bonnet et al., 2021), transparent and replicable (Cendrero et al., 2003), and should quantify uncertainties in outcomes (Uwizeye et al., 2017). While some of these concerns have been addressed in other work, for example, in the design of the SDG-index dashboard (Schmidt-Traub et al., 2017), we lack frameworks to understand how normative choices in index design tilt their outcomes.

This work (a) demonstrates how well-known indices differ sharply in their assessment of countries' climate performance; (b) highlights how normative choices in their design lead to contrasting results; (c) develops a framework to assess the degree to which indices embody different worldview; and (d) shows how indices' outcomes can be understood based on this framework.

We begin by comparing three prominent climate performance indices: the Climate Change Performance Index (CCPI) (Burck et al., 2023), the Environmental Performance Index (EPI) (Block et al., 2024; Wolf et al., 2022b), and the Climate Action Tracker (CAT) (CAT, 2023).

We then focus on the EPI, which has established itself as a comprehensive, data-driven, replicable measure of environmental performance (Hsu & Zomer, 2016), assessing 180 countries on climate change mitigation, environmental health, and ecosystem vitality. While widely recognized for offering a holistic perspective on environmental performance, as all composite indices, the EPI's results are influenced by methodological choices (Pinar, 2022), including indicator selection, data imputation, weighting, aggregation, and normalization methods (Greco et al., 2019; Libório et al., 2022; Mazziotta & Pareto, 2022). To address these concerns, the EPI's methodology and results undergo periodic audits and tests for data and statistical validity (Smallenbroek et al., 2023). These audits revealed that the EPI results are sensitive to the weight attributed to different sets of indicators (Pinar, 2022; Smallenbroek et al., 2023). This highlights how the outcomes of composite indicators widely cited in the media, and potentially influencing policymaking, may be influenced by their methodology.

Such concerns were behind the extensive debate in the Indian media after the country ranked at the bottom of the EPI-2022 (e.g. Bhushan, 2022; Dubash & Lele, 2022). The Indian *Ministry of Environment, Forest and Climate Change* challenged the EPI's findings, claiming that they were based on unfounded assumptions and unscientific methods (PIB, 2022), to which the EPI issued a rebuttal (Wolf et al., 2022a). Similarly, in 2019, the CCPI issued a rebuttal to the then-Australian Prime Minister's claim that the CCPI's 2019 index lacked credibility for classifying Australia as a poor performer (Germanwatch, 2019).

How does the ostensibly objective task of gauging climate performance get mired in such contentious debates? Prior research has shed little light onto this question, which motivated this work. Here, we demonstrate that the divergence between well-known climate performance indices partly arises from differences in their emphasis on equity. We posit that these divergences in outcomes, and the associated real-world impacts, can emerge from contrast in the normative choices embodied by the index's methodology.

The paper is organized as follows: Section 2 outlines the construction of an equity-focused version of the EPI (EPI-equity), and the conceptual framework we develop to quantify the role of equity in performance indices. Results are discussed in Section 3, with their implications for index developers, policymakers, the media, and civil society discussed in Section 4. We close the paper with a summary and outlook in Section 5.

2. Data and methods

2.1. Methods

To show the influence of normative assumptions on climate indices' results, we (1) analysed three climate performance indices: EPI, CCPI, and CAT, (2) developed an equity-focus version of the EPI-2022, called the EPI-

equity, and (3) created a conceptual framework to classify indices based on the principle of equity and climate justice (see SI for detailed methodology).

2.1.1. Environment performance index (EPI)

Two version of the EPI, i.e. EPI-2022 and EPI-2024 (Block et al., 2024; Wolf et al., 2022b), are analysed, focussing only on their climate change policy objective.

2.1.2. The equity-based EPI (EPI-equity)

We develop EPI-equity to illustrate the role of equity in performance assessment. Conceptually, the EPI-equity reflects a 'national emissions fair share' approach based on the principle of equal per capita emission rights (Matthews, 2016). It is a modification of EPI-2022 (Wolf et al., 2022c) since the Technical Appendix of EPI-2024 was not available at the time of analysis. The EPI-equity is identical to EPI-2022s climate change objective, except for two indicators that together account for 72% of the weight: *Projected 2050 GHG Emissions* (GHN), and *the adjusted CO_2 growth rate over 2010-2019* (CDA).

GHN was modified to 'EQA', scoring countries based on whether they remain within their *fair share* of the global cumulative GHG *emissions* from 1850 to 2050. A country's 'fair share' is defined as the global cumulative emissions from 1850 to 2050 multiplied by the country's mean share of the world's population over that period. Following the EPI, future emissions (2020 to 2050) are computed by linearly projecting the 2010–2019 emissions trend. Countries remaining within their allocated fair share were scored from 0-100 based on the *remaining share*, and those exceeding their fair share were scored 0.

CDA was modified to 'EQI,' the *rate of depletion* of each country's allocated share of CO_2 emissions based on cumulative global emissions between 2010 and 2019. Countries remaining within their allocated share are scored from 0-100 based on the *rate of depletion*, and countries exceeding their fair share were scored 0.

Since the EPI-equity differs from EPI-2022 in only two modifications, the change in score can be expressed as:

$$\textit{Score}\ \text{EPI-equity} - \textit{Score}\ \text{EPI-2022} = 0.36\ (\textit{EQA} - \textit{GHN}) + 0.36\ (\textit{EQI} - \textit{CDA}), \tag{1}$$

where 0.36 is the weight for CDA and GHN each in EPI-2022.

The modified terms – EQA and EQI – can be interpreted as a 'pure equity' approach since they are defined based only on countries' shares of total emissions. They differ from 'remaining carbon budgets' associated with specific temperature targets (e.g. 1.5 °C). This unconventional choice was made for simplicity and alignment with EPI-2022s approach.

EPI-equity is not intended as a proposal for a new performance index but as an analytic tool for illustrative purposes.

2.1.3. Conceptual framework to identify normative choices

We develop a conceptual framework to analyse climate indices based on two normative choices: the physical climate approach, and the equity approach.

The physical climate approach assesses countries *contribution to climate change,* and is therefore concerned with absolute emissions. In contrast, the equity approach assesses country's performance from a *socio-economic perspective* considering factors such as population or Gross Domestic Product (GDP). Here, the equity approach is operationalized by considering per capita emissions and fair shares of global cumulative emissions.

The framework is a scoring system that positions indices in a 'space of normative choices' spanned by the climate and equity axes. Each indicator in an index is ascribed a score of 1-3 based on its alignment with the criteria defined for each axis (summarized in Figure 4(a) with details in Sec. S1.4 and Table S3). Indices' position on this space has important implications for their interpretation (see Sec 3.4 for further details).

2.2. Data

EPI-equity uses the same GHG emission dataset used by EPI-2022, covering 180 countries from 1850 to 2019, and originally sourced from the PRIMAP-hist dataset (Gütschow et al., 2016). Historical and projected

population data, projected till 2050, come from Our World In Data portal, originally sourced from UN World Population Prospects (UN WPP, 2024) (details in SI; section S1).

Also analysed are CCPI-2024, which covers 63 countries (Burck et al., 2023), and CAT, which covers 39 (CAT, 2023). Since limiting the analysis to the 26 countries rated by all three indices (nine developed, 17 developing) yields too small a sample size, pairwise intersections are used: Figure 1(a) includes 35 countries common to EPI and CCPI, and Figure 1(c) includes 58 countries common to CCPI and CAT (See Table S1). Since the time of our analyses, the CCPI-2025 (Burck et al., 2024) was published with only minor changes in countries' performance (Fig. S1) that do not change the conclusions of this study.

All datasets were accessed on August 2024 and Table S2 shows their descriptive statistics.

3. Results

3.1. Comparison of climate indices

We compared the climate performance of countries on EPI-2024, CCPI-2024, and CAT (Figure 1). Countries rated 'critically insufficient,' 'highly insufficient', and 'insufficient' by CAT exhibit progressively higher average EPI-2024 scores. However, this trend brakes with the highest CAT rating of 'almost sufficient,' whose average

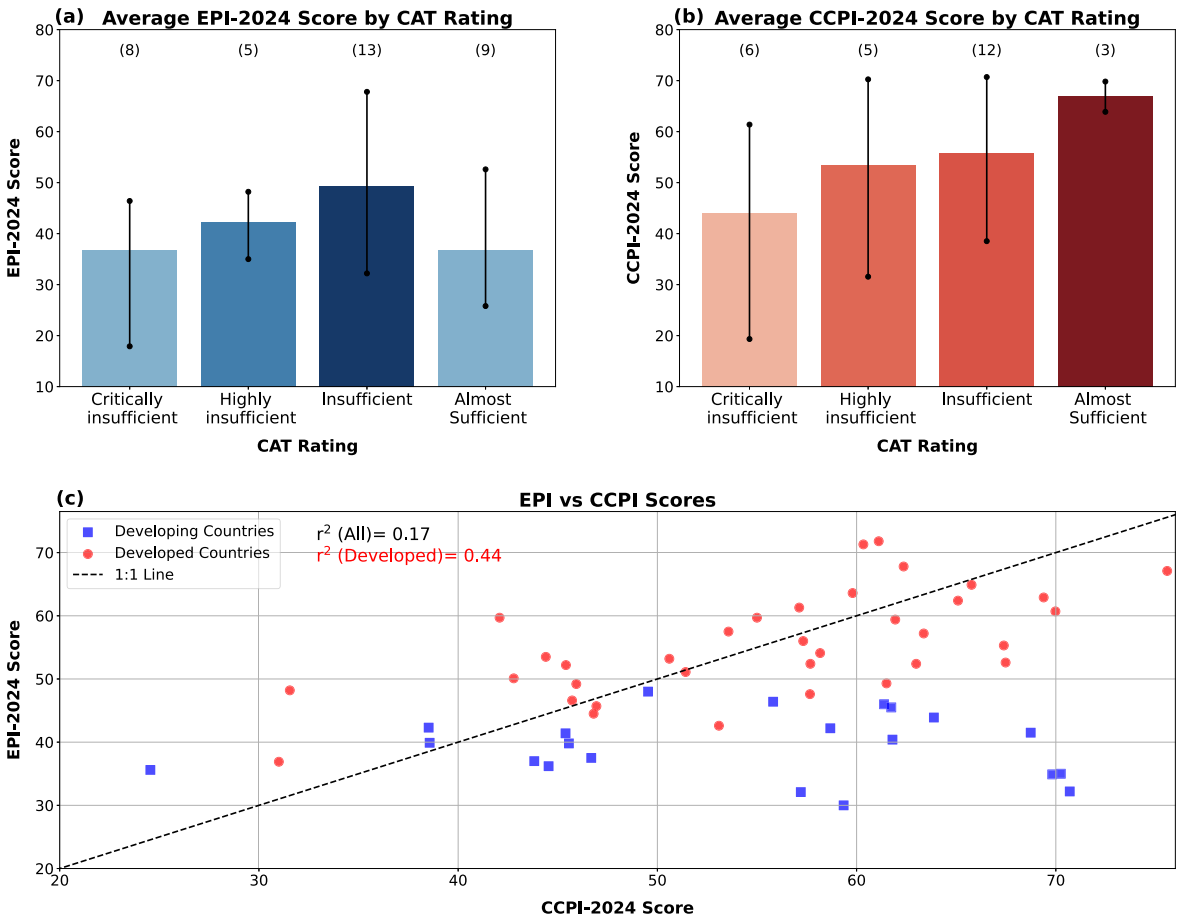

Figure 1. Comparison of three climate performance indices. (a) shows the average EPI-2024 score for countries in each CAT category, with the full range represented by black lines. (b) is identical to (a) for the CCPI-2024 scores. The number of countries contributing to each bar is indicated above it in parentheses. (c) is a scatter plot of EPI-2024 and CCPI-2024 scores, categorized as Developing (blue squares) or Developed (red circles) following UNDP classification. A dashed black line represents the 1:1 line. Higher scores indicate better performance.

score on EPI is close to the 'critically insufficient' group's score. In contrast, the average CCPI-2024 score progressively rises for each higher-rated CAT group (Figure 1(b)), showing more consistency with CAT's score than the EPI.

While on the EPI higher scores are occupied predominantly by developed countries, both developing and developed countries attain high scores (> 60) on the CCPI (Figure 1(c)). Countries with the highest departure from the 1:1 line in Figure 1(c) are exclusively developing countries. The EPI tends to assess developing countries' climate performance more poorly than the CCPI and CAT.

3.2. Role of equity in performance assessments

To illustrate the impact of internalizing equity in climate performance assessments, EPI-2022/2024 was compared to EPI-equity, revealing distinctive geographical patterns in performance on these three metrics (Figure 2).

In the EPI-2022/2024, Europe tends to perform well, whereas the poorest performers are in Africa, the Middle East, and South and Southeast Asia (Figure 2(a,b)). The Americas, China, and Russia occupy intermediate ranks. This aligns with the relatively poor performance of developing countries on the EPI observed in Figure 1(b).

Performance on the EPI-equity exhibits contrasting patterns (Figure 2(c)). Most top-performing countries on EPI-2024 perform poorly on EPI-equity, and vice versa. On aggregate, the Global South benefits under the equity approach, while the Global North's performance declines. This analysis demonstrates how incorporating equity into a climate performance metric can significantly influence outcomes.

To further diagnose the disparities in performance on EPI and EPI-equity, an analysisof Eq. 1 is useful. The first term on the right-hand side ($EQA - GHN$) represents the anomaly arising from the difference in scores based on the exceedance of a country's allocated fair share of the global 1850–2050 GHG emissions relative to the score based on its absolute GHG emissions in 2050. The second term ($EQI - CDA$) is the anomaly arising from the difference in scores based on the rate of change in a country's allocated fair share of the 2010-2019 global CO_2 emissions relative to the score based on the absolute rate of change in its CO_2 emissions over that decade. These two terms are plotted in Supplementary Fig. S2.

For Global North countries, score anomalies on the first term of Eq. 1 are generally small (Fig. S2a), since they score poorly on both GHN and EQA due to high 2050 projected emissions that exceed their allocated fair share of cumulative emissions. Conversely, many Global South countries exhibit positive anomalies on the first term since their projected cumulative emissions (1850–2050) remained within their allocated fair share.

A more pronounced contrast emerges between the Global North and Global South on the second term of Eq. 1 (Fig. S2b). Here too, Global South countries feature positive anomalies, since they remain within their decadal fair share allocations despite rising emissions. Concurrently, the Global North features substantial negative anomalies, since the EPI rewards countries achieving absolute decoupling, which are mainly Global North countries (Pao & Chen, 2019; Wu et al., 2018). The decline in their scores under equity considerations indicates that these countries, despite achieving emissions decoupling in the past decade, have nevertheless exceeded their decadal fair share allocation.

Notably, the performance of some of the EPI top scoring countries (Denmark, UK, Finland, and Sweden) declines prominently on *both* terms of Eq. 1. China's performance also declined on EPI-equity relative to EPI, suggesting it is now closer to Global North countries than to the Global South (Hickel, 2020).

This analysis illustrates how normative choices and conceptual frameworks can shape climate indices' outcomes and associated narratives (Greco et al., 2019). While this is illustrated with the EPI, these conclusions should hold widely across climate performance indices.

3.3. Aggregate performance by region and developmental status

Here, we look beyond individual countries to groups classified by development status and region. Figure 3(a) classifies countries by development status following the United Nations Development Programme (UNDP, 2022). Performance on EPI-2022 and EPI-2024 is similar across the three development categories. However, a pronounced difference emerges between the EPI-equity and the original EPI metrics for the 'Developed' and

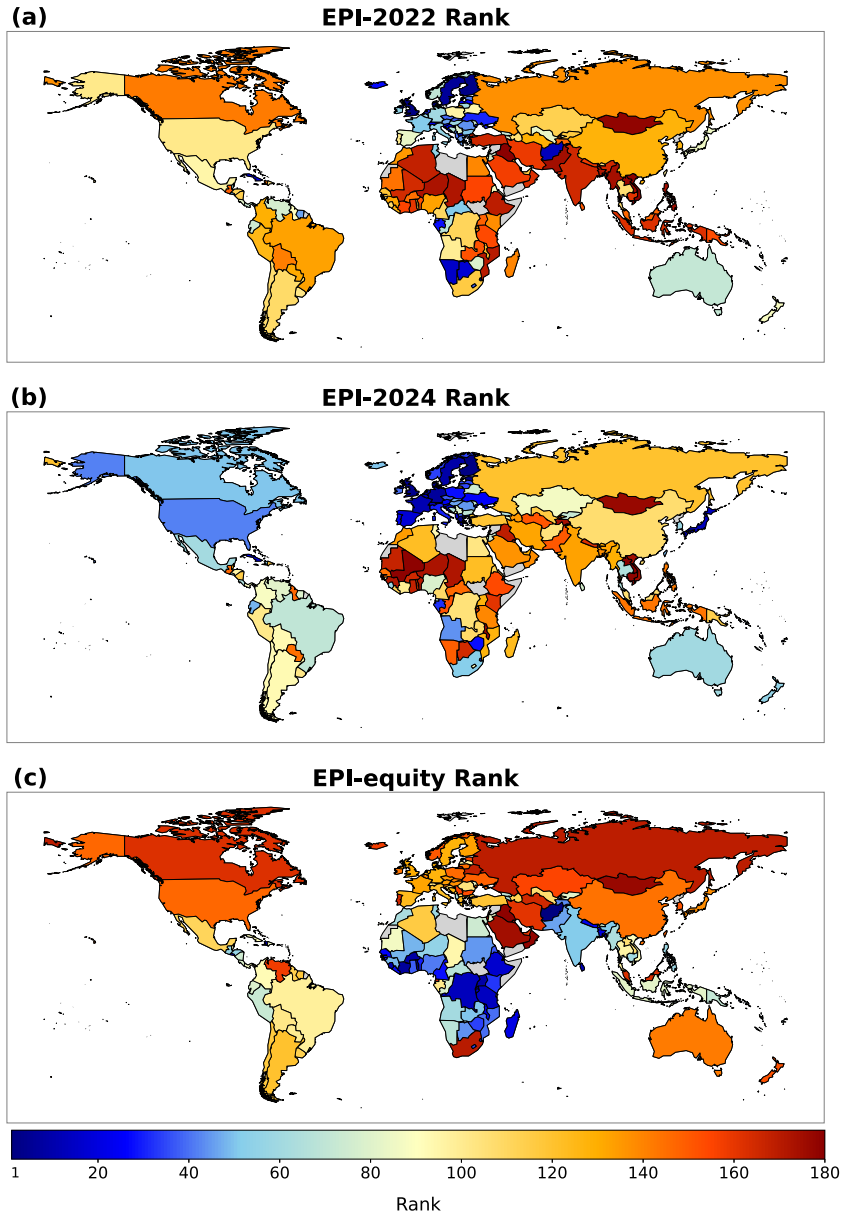

Figure 2. Spatial variation in ranks on (a) EPI-2022, (b) EPI-2024, and (c) EPI-equity. Lower ranks (blues) indicate better performance. Countries without data are shaded in grey. Political boundaries may not be representative of all viewpoints.

'Least Developed' groups. The 'Developed' group performs well on the original EPI metrics while it drops sharply on EPI-equity. The decline in performance of the Developed group on EPI-equity is largely the result of the rapid rate of depletion of their 2010-2019 fair share allocation due to insufficient rates of emissions reductions (second term of Eq. 1, Fig. S2).

The performance of the Least Developed group exhibits the opposite trend. This group performs better on EPI-equity because their emissions remain within fair share allocations, despite projected emissions increases.

Consistent with the patterns across development status, the Global North, Eastern European, South Asian, and Sub-Saharan African regions exhibit the most conspicuous difference in performance between the EPI-equity and the original EPI metrics (Figure 3(b)). The Global North and Eastern Europe perform well on EPI-

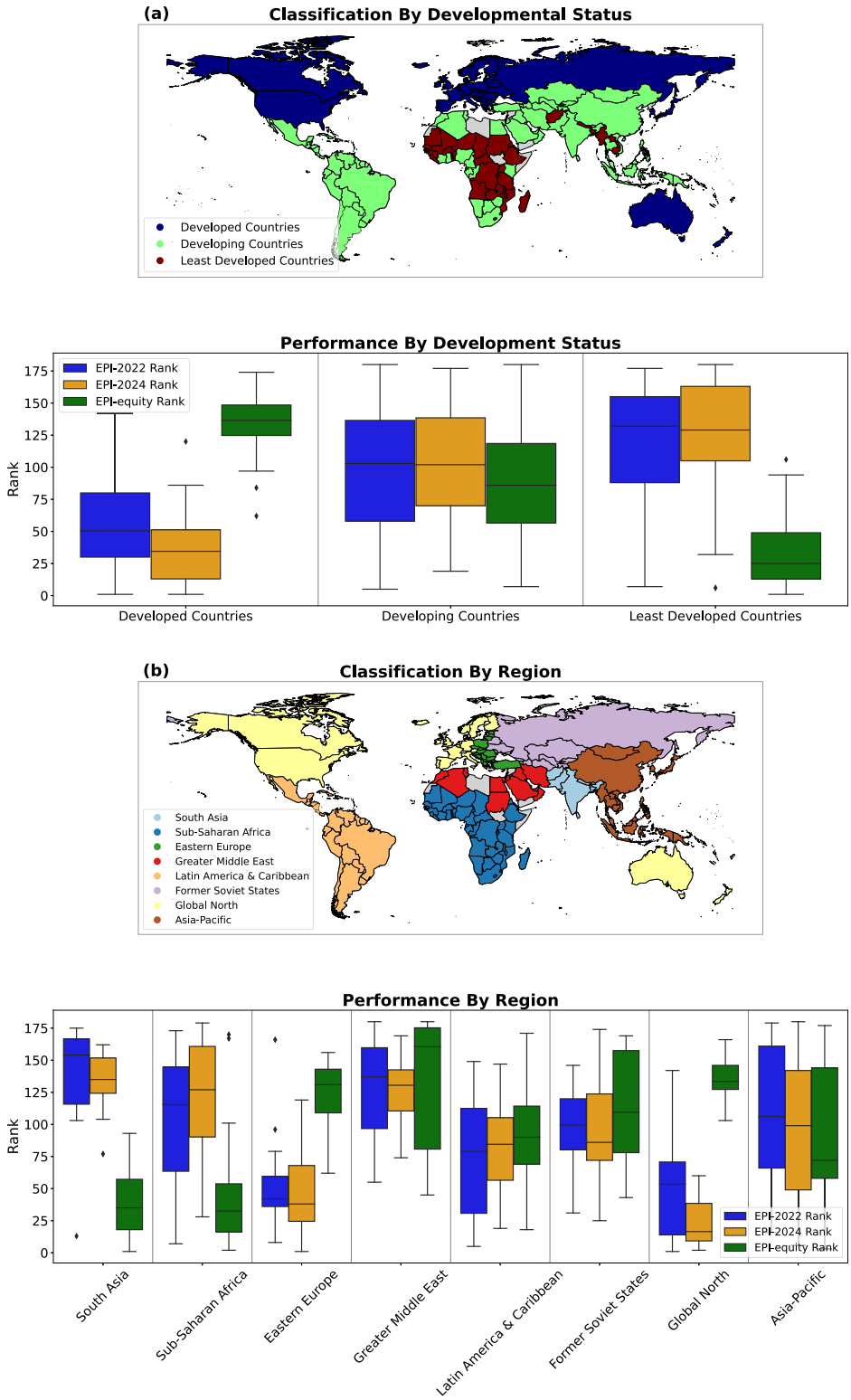

Figure 3. Performance on EPI-2022, EPI-2024, and EPI-equity categorized by (a) development status, and (b) region. The maps in (a) and (b) classify countries in their respective categories, and boxplots represent their ranks on the three metrics. Boxes span the interquartile range and the central lines in each box represent the median. Whiskers denote 1.5 x interquartile range. Lower ranks indicate better performance. Countries without data are shaded in grey. Political boundaries may not be representative of all viewpoints.

2022/2024, whereas South Asia and Sub-Saharan Africa perform poorly. Under EPI-equity, these patterns are reversed. Notably, the contrast in the performance of the 'Global North' group is even more pronounced compared to the 'Developed' group of countries, since the former is a subset of the latter (see Figure 3).

In summary, based on development criteria, the 'Least Developed' group benefits most by being assessed from the equity approach, whereas the 'Developed' group is most disadvantaged. In regional terms, South Asia and Sub-Saharan Africa benefit most from the equity approach, whereas the Global North and Eastern Europe are most disadvantaged.

While EPI-equity is presented as an illustrative tool, its results resonate with real-world debates. Countries of the Global South rise in rank, reflecting their equity-based perspectives in UNFCCC negotiations (United Nations, 1992), while developed nations fall, aligned with the critique of their overuse of the atmospheric commons (Hickel, 2020). This suggests that even a simplified equity lens can capture key political fault lines.

3.4. Conceptual framework to identify normative choices embodied in climate performance indices

Indicators inevitably encode worldviews, often imperceptibly (Meadows, 1998). The preceding analysis demonstrated that divergent underlying worldviews can result in contrasting outcomes. Thus, identifying broad principles underpinning indices can help understand their disparities. To this end, we develop an illustrative framework to interrogate climate indices on the normative choices they embody.

The framework is well motivated with the example of Indonesia. From a cumulative GHG emissions approach, Indonesia is considered one of the largest contributors to climate change (Jones et al., 2023). However, from the viewpoint of national fair shares based on per capita emissions, Indonesia has contributed substantially towards *avoiding* climate breakdown (Hickel, 2020).

This illuminates two contrasting approaches, which allows us to discern two distinct normative choices embodied in climate performance indices. The first is an exclusively biophysical approach, termed 'physical climate approach' since global warming is proportional to the accumulated CO_2 in the atmosphere (Allen et al., 2009; Matthews et al., 2009). This approach bears analogy to the 'polluter pays' principle applied at the country scale (Tilton, 2016), and assesses cumulative GHG emissions. This approach is indifferent to socio-economic concerns such as the distinction between 'luxury emissions' of the rich and 'survival emissions' of the poor (Agarwal & Narain, 1991). The second is a socio-economic approach, termed the 'equity approach', and grounded in the principle of equitable access to the atmospheric commons of every individual through per capita emissions (Ghosh, 1993; Grübler & Fujii, 1991; Neumayer, 2000; Smith, 1991). These two approaches, each justified by established scientific and ethical principles, lead to strikingly different interpretations of Indonesia's climate performance.

We construct the framework based on this conceptual division. Along the 'physical climate' axis, indicators based on *cumulative emissions* are assigned a score of 3, and those focused on *annual emissions with* and *without* reference to an emissions pathway are ascribed scores of 2 and 1, respectively (Figure 4(a)).

Along the 'equity' axis, scores of 3, 2, and 1 are assigned to indicators based on *fairshare emissions*, *cumulative per capita emissions*, and *annual per capita emissions*, respectively. These choices were informed by an analysis of the indicators comprising the EPI, CCPI, and CAT (and their variants). Details are provided in the SI (Table S3).

Applying the framework to climate indices reveals significant differences in their representation of each normative choice (Figure 4(b)). Table S4 provides calculation details. EPI-2022 is the metric most strongly aligned with the physical climate approach, albeit with minimal representation of the equity approach. Its successor, EPI-2024, exhibits a decreased emphasis on physical climate counterbalanced by a modest increase in its equity representation. This shift towards equity is limited since equity-based pilot indicators carry only 5% weight on EPI-2024s climate change objective (Block et al., 2024). In contrast, the CCPI focuses exclusively on per capita emissions, thereby aligning solely with the equity approach. The CAT demonstrates a more balanced approach, representing both approaches more robustly than EPI-2024, although neither to high degree.

Our analysis demonstrates that the stronger correlation between CCPI and CAT ratings, compared to their relationship with EPI (Figure 1), can be explained by their shared emphasis on equity (Figure 4). This suggests

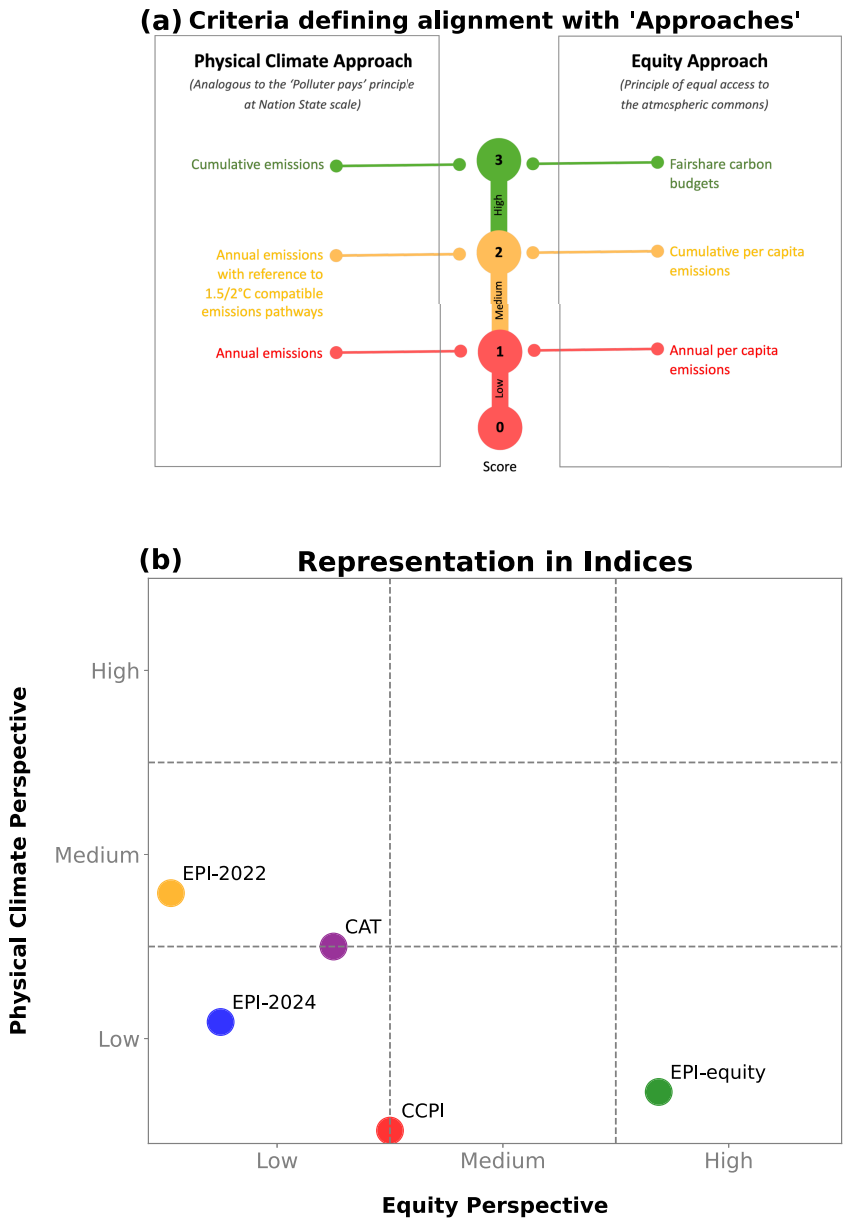

Figure 4. (a) summarizes the operationalization of the 'perspectives framework' listing the three criteria chosen to represent each dimension ('physical climate' and 'equity'). Discrete scores of 1-3 are associated with each criterion. (b) is a categorization of multiple climate indices based on their representation of each dimension. The vertical axis denotes the 'physical climate approach' and the horizontal axis denotes the 'equity approach'. The score of an index on a given dimension is the weighted sum of the discrete scores specified in (a), as described in the SI. A score between 0 and 1 is qualified as 'Low,' 1 and 2 as 'Medium', and 2 and 3 as 'High.' Refer to the SI for details.

that apparent contradictions between climate indices may stem from differences in the normative choices they embody rather than methodological or data choices alone.

The EPI-equity stands out by exhibiting the strongest representation of the equity approach, primarily due to its use of fair share allocations in performance assessment. This enhanced focus on equity comes at the expense of physical climate representation.

We note that the equity scoring system developed here addresses only a part of the larger equity issue. More broadly, approaches to equitable burden sharing, including the UNFCCC's 'Common but differentiated

responsibilities and respective capabilities (CBDR-RC)' approach are based on four core principles: Responsibility, Equality, Capability, and the Right to Development (Fleurbaey et al., 2014; United Nations, 1992). The latter three are inherently socio-economic, thus equity-centric. Of these, 'equality' appears directly in our scoring system, via per capita metrics. However, 'Capability' and 'Right to Development' do not appear because none of the target indices (EPI, CCPI, CAT) explicitly incorporates these aspects of equity (details in Table S4).

The Responsibility principle is an instructive case study. Our framework conceives post-1850 cumulative emissions – often thought of as quantifying 'responsibility' – as the strongest representation of physical climate, but without mention in relation to equity (see Figure 4(a)). However, as noted above, the concept of responsibility is also legitimately seen as representing a conception of equity. This might seem at odds with our framework at first glance. The fundamental concept underpinning the 'physical climate' axis is *contribution to* climate change, which is fundamentally different from *responsibility for* climate change (Müller et al., 2009). The former refers to *causal* responsibility, i.e. the fact that current climate is a consequence of humanity's historic cumulative emissions (Arias et al., 2021; Box TS.1). The latter refers to *moral* responsibility, i.e. ethical accountability considering factors such as knowledge, intent, and agency (Müller et al., 2009).

Our framework engages with *causal* responsibility, which is unambiguously associated with the physical climate dimension. The equity axis in the framework does not have a scoring element associated with *moral* responsibility, since such a term does not appear in the indices considered here. An indicator can conceivably operationalize *moral* responsibility by choosing a later start date (e.g. 1960 or 1990) for counting cumulative emissions – this can partially account for issues such as climate knowledge or intent. This highlights the 'one-dimensional' nature of the framework, in which an indicator is necessarily mapped onto just one axis for simplicity and ease of interpretation. However, indicators may plausibly encompass both approaches, such as ones focused on 'responsibility'. Such indicators may warrant exploring two-dimensional expansions of our formulation that can map indicators onto both axes. The scoring system presented here is tailored to the three indices being analysed, rather than to the full range of climate performance indices or equity conceptions. It is a proof-of-concept to highlight the utility of such frameworks for interpretation. While adjustments to the scoring criteria would likely alter the relative positions of indices in Figure 4(b), the fundamental insight that indices inherently embody multiple choices, and to different degrees, would remain robust.

4. Policy-Relevant interpretations and implications

Analysis of three prominent indices revealed disparities in evaluating countries' performance, demonstrating that differing emphases on equity can strongly shape outcomes. Our conceptual framework illustrates that embodying certain normative choices more strongly than others tends to benefit specific groups of countries.

While measuring countries' absolute cumulative emissions offers the best picture of their contributions to climate change, this tends to penalize countries with large economies and populations while favouring smaller countries. In contrast, a sole focus on per capita emissions is also inadequate, as it can conceal steep increases in absolute emissions and the attendant urgency of mitigation. This is a challenging dichotomy; while both the cumulative and per capita emissions have compelling arguments in their favour, neither approach adequately addresses the problem of climate change in isolation. This paper argues that adopting one normative choice *to the exclusion of the other* risks tilting the emergent narrative.

Despite their allure of objectivity, data-driven indices are shaped by methodological normative choices. The impacts of indicator selection, weighting, and aggregation methods on indices' results are well recognized (Booysen, 2002; Böhringer & Jochem, 2007; Greco et al., 2019; Singh et al., 2012). In line with our results, Greco et al. (2019) recognized that the normative selection of indicators and weights fundamentally alters the conclusions derived from a composite indicator. We expand on this insight by proposing an interpretative lens to probe an index's underlying value system (Meadows, 1998). Deeper and clearer awareness of the value systems underlying climate indices can facilitate their interpretation and lead to the development of more robust, comprehensive, and useful analyses of climate performance.

The results presented here show that different climate indices may be valid depending on the context and purpose. However, clarity and transparency about the subjective underpinnings of indicators is always critical.

Without it, politically motivated agents can pick the indices that better support their worldview and use them as rhetorical instruments. For instance, our results suggest that the Indian and Australian governments' rejections of the EPI and CCPI, respectively, were driven by differing normative priorities. The Indian government's stance was an articulation of the need for equity in performance assessments, at odds with the EPI's approach, while the Australian government's response amounted to the rejection of the CCPI's predominant focus on equity.

Based on these observations, we make the following recommendations for policy.

4.1. Implications for index developers

Index developers must recognize that assessing climate performance cannot be a value-free exercise based solely on data. Choosing which data to assess, and against which benchmarks, are crucial judgements that tilt outcomes and shape narratives (Uhlich, 2023). We propose that:

- Index documentation should clearly articulate both the methods *and* the normative choices underpinning the index, rather than presenting rankings as objective assessments.
- Developers could provide interactive tools allowing analysts explore whether, and how, performance assessments change when they emphasize different normative choices. Modular indices with separable components that can be recombined for different purposes may be considered.
- Index audits, such as Smallenbroek et al. (2023), should incorporate tests of normative choices to help developers understand whether their index aligns with stated objectives or inadvertently promotes particular worldviews. The goal is not methodological neutrality but conscious transparency.
- Developers may consider making explicit comparisons to other popular indices that take contrasting viewpoints, illustrating by example how these viewpoints explain contrasting results.

4.2. Implications for policymakers, media, and civil society

- Understand indices' methodological details and normative choices before citing their rankings.
- Avoid relying on a single index for comprehensive climate performance evaluation. Contextualize the outcomes of an index with other credible ones that offer a complementary narrative.
- Policymakers should transparently specify why particular indices are appropriate for their decision-making contexts, recognizing that equity-focused indices may be more relevant for climate justice while absolute emissions-focused indices may better inform mitigation.
- More constructive debates and action may result from contextualizing indices based on recognizing their embedded choices. Even indices emphasizing a singular viewpoint may highlight genuine policy gaps that warrant attention.

5. Discussion and conclusions

Our analysis demonstrates that multiple perspectives, seemingly irreconcilable, can co-exist for complex social-environmental problems, each justified by well-established scientific and ethical principles. Prioritizing one approach over others risks conflicts arising from fragmented worldviews. However, considering multiple indices sampled from the 'space of normative choices' can reduce the likelihood of such conflicts by ensuring a balanced representation of viewpoints.

Our framework is not comprehensive and the binary division of normative choices is an oversimplification. We chose simplicity and parsimony for illustrative purposes. While there are many environmental equity and fairness principles (Adger et al., 2006; Höhne et al., 2014; Kanitkar & Jayaraman, 2019; Muttitt & Kartha, 2020; van den Berg et al., 2020; Wallimann-Helmer et al., 2019), the focus here was on principles most pertinent to the three indices analysed.

Future research may add dimensions to this framework when necessary for specific contexts. For instance, political feasibility (Montfort et al., 2025), moral responsibility (Müller et al., 2009), and capability (Jewell & Cherp, 2020)

are alternative dimensions that may be pertinent to performance assessments. Embodied choices may even be related to the intended *purpose* of an index. For example, the low scores of EPI-2024 on both the climate and equity suggest that the index prioritized a different dimension (i.e. exemplary policy success). However, to remain useful and insightful, expansions of the framework must continue to strive to for parsimony.

The choice of criteria in the framework we present is meant to identify normative choices in index development, but is *itself* normative, reflecting our view on equity. For example, this approach rewards 'fair share budgets' but disregards 'grandfathering' since we view the latter as incompatible with the central equity tenet of protecting the most vulnerable (Dooley et al., 2021; Moellendorf, 2020). Ascribing a higher score to 'fair share budgets' than to 'cumulative per capita emissions' on the equity axis also reflects a normative view about 'strong fairness' principles that can be disputed from a different normative standpoint. Similarly, other equity principles (capability, responsibility) are likely to be more pertinent in other contexts. Thus, the scores reflect our viewpoint about what constitutes stronger alignment with a particular approach, not objective truths about methodological superiority.

It is important to also note that not all methodological choices are normative. An example of 'pure' methodological choices (independent of normative choices) may be choice of aggregation using arithmetic or geometric mean. This contrasts with 'loaded' methodological choices in which normative choices are implicit. We focused on exploring the implications of 'loaded' methodological choices that might otherwise be construed as a value-free exercise. The design of EPI-equity deliberately maintained the mathematical structure of EPI-2022 while modifying two of its indicators in order to isolate and showcase the impact of normative methodological choices. This study, however, does not explore the precise distinction between 'pure' methodological choices versus those with normative implications, which is an important limitation of the work presented here and could be an area for future exploration (Sec. S1.5).

Our framework's value lies in revealing hidden assumptions made in the development of climate indices. Making these assumptions transparent facilitates a more informed debate about which approaches are most appropriate for specific purposes. We expect that this approach will be built upon in future work, perhaps as a question-based rubric that interrogates indices on multiple equity concepts. The goal is to develop tailored formalisms that can discern values, assumptions, and worldviews embedded in a range of environmental indices towards facilitating enhanced transparency, fostering dialogue between differing viewpoints, and eventually leading to more thoughtfully designed indices to guide policy (Meadows, 1998).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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